WHAT IS CLAIMED:

1. A method for determining a glycan structure on a glycosylated ~~molecule~~protein, the method comprising:

obtaining, for each of a plurality of bioreactor runs comprising culturing cells to produce the glycosylated protein, levels for one or more glycan structures on the glycosylated ~~molecule~~ protein using a process analytical technology (PAT) tool, wherein the PAT tool utilizes or otherwise comprises Raman spectroscopy, wherein the obtaining is within one or more first bioreactors having a first volume equal to or below a first threshold, the PAT tool obtaining spectral data;

generating one or more regression models based on the obtained spectral data that correlate levels of the one or more glycan structures on the glycosylated ~~molecule~~protein with the obtained spectral data;

measuring the one or more glycan structures on the glycosylated ~~molecule~~protein using the PAT tool, wherein the measuring is within one or more second bioreactors having a second volume equal to or above a second threshold to result in measured spectral data; and

determining, by at least one computing device using the generated one or more regression models and based on the measured spectral data, levels of the one or more glycan structures on the glycosylated ~~molecule~~protein within one or more second bioreactors.

2. The method of claim 1:

(a) further comprising refining the one or more regression models based on the combination of the obtained spectral data and the measured spectral data;

(b) further comprising:

(i) maintaining one or more operating parameters of the one or more second bioreactors based on the determined levels to produce a desired glycosylated ~~molecule~~protein; or

(ii) selectively modifying one or more operating parameters of the one or more second bioreactors based on the determined levels to produce a desired glycosylated ~~molecule~~protein,

wherein the one or more operating parameters optionally comprise a pH level, a nutrient level, a culture media concentration, a frequency interval for culture media addition, or a combination thereof,

wherein the nutrient level is optionally selected from the group consisting of: a concentration of glucose, a concentration of lactate, a concentration of glutamine, and a concentration of ammonium ions, and

wherein the concentration of glucose is optionally automatically modified based on the measured spectral data; and/or

(c) further comprising purifying the glycosylated ~~molecule~~protein.

3. The method of claims 1 or 2, wherein:

(a) the glycan structure is selected from the group consisting of: G0F-GlcNac, G0, G0F, G1F, and G2F, or a combination thereof; and/or

(b) the obtaining is within two or more bioreactors having different volumes.

4. The method of any one of claims 1 to 3, wherein

(a) the first threshold is:

(i) about 250 liters or less;

(ii) about 100 liters or less;

(iii) about 50 liters or less;

(iv) about 25 liters or less;

(v) about 10 liters or less;

(vi) about 5 liters or less;

(vii) about 2 liters or less;

(viii) about 1 liter or less; and/or

(b) the second threshold is:

(i) about 1,000 liters or more;

(ii) about 2,000 liters or more;

(iii) about 5,000 liters or more;

(iv) about 10,000 liters or more;

(v) about 15,000 liters or more;

(vi) about 10,000 to about 25,000 liters

(vii) at least 5X greater than the first threshold;

(viii) at least 10X greater than the first threshold;

(ix) at least 100X greater than the first threshold; or

(x) at least 500X greater than the first threshold.

5. The method of any one of claims 1 to 4, wherein

(a) the first volume is:

(i) about 0.5 to about 250 liters;

(ii) about 1 to about 50 liters;

(iii) about 1 to about 25 liters;

(iv) about 1 to about 10 liters; or

(v) about 1 to about 5 liters; and/or

(b) the second volume is:

(i) about 1,000 to about 25,000 liters;

(ii) about 2,000 to about 25,000 liters;

(iii) about 5,000 to about 25,000 liters;

(iv) about 10,000 to about 25,000 liters; or

(v) about 15,000 to about 25,000 liters.

6. The method of any one of claims 1 to 5, wherein:

~~(a) the PAT tool utilizes or otherwise comprises Raman spectroscopy;~~

(a~~b~~) the one or more regression models comprise a partial least squares (PLS) model;

(b~~e~~) the glycosylated ~~molecule~~protein comprises a monoclonal antibody (mAb) or a non-mAb;

(c~~d~~) the determining step is performed

(i) on-site or off-site; and/or

(ii) in-line, at-line, on-line, off-line, or a combination thereof;

(d~~e~~) the obtaining comprises: receiving data characterizing the spectral data from the PAT tool; and/or

(e~~f~~) the generating is executed by one or more computing devices.

7. A method of producing a glycosylated ~~molecule~~protein having a desired glycan structure, the method comprising:

measuring one or more glycan structures using a process analytical technology (PAT) tool to result in spectral data, wherein the PAT tool utilizes or otherwise comprises Raman spectroscopy, wherein the measuring is within a bioreactor having a volume of equal to or above 1,000 liters;

determining, by at least one computing device using one or more regression models and based on the measured spectral data, a level of the one or more glycan structures within the bioreactor, wherein the one or more regression models are generated using trial runs comprising culturing cells to produce the glycosylated protein from at least one bioreactor

having a volume less than or equal to 50 liters and at least one bioreactor having a volume equal to or above 1,000 liters; and

maintaining one or more operating parameters of the bioreactor when:

the level of a desired glycan structure is above a pre-defined threshold, or

the level of an undesired glycan structure is below a pre-defined threshold; and

selectively modifying one or more operating parameters of the bioreactor when:

the level of a desired glycan structures is below a pre-defined threshold, or

the level of an undesired glycan structure is above a pre-defined threshold.

8. The method of claim 7, wherein:

(a) measuring is performed in-line, at-line, on-line, off-line, or a combination thereof;

(b) measuring occurs:

(i) more than once daily;

(ii) about every 5 to 60 minutes;

(iii) about every 10 to 30 minutes;

(iv) about every 10 to 20 minutes; or

(v) about every 12.5 minutes;

(c) measuring is within a bioreactor having a volume of:

(i) about 2,000 liters or more;

(ii) about 5,000 liters or more;

(iii) about 10,000 liters or more;

(iv) about 15,000 liters or more;

(v) about 10,000 liters to about 25,000 liters; or

(vi) about 15,000 liters;

(d) the determining step is performed on-site or off-site; and/or

(e) the desired glycan structure is selected from the group consisting of: G0F-GlcNac, G0, G0F, G1F, and G2F, or a combination thereof.

9. The method of claim 7 or 8, wherein:

(a) the bioreactor is a batch reactor, a fed-batch reactor, a perfusion reactor, or a combination thereof;

(b) the one or more operating parameters comprise a pH level, a nutrient level, a culture media concentration, a frequency interval for culture media addition, or a combination thereof,

wherein the nutrient level is optionally selected from the group consisting of: a concentration of glucose, a concentration of lactate, a concentration of glutamine, and a concentration of ammonium ions; and

wherein the concentration of glucose is optionally automatically modified based on the measured spectral data; and/or

~~(c) the PAT tool utilizes or otherwise comprises Raman spectroscopy; and/or~~

(c~~d~~) the one or more regression models comprise a partial least squares (PLS) model.

10. A system for producing one or more glycosylated ~~molecules~~proteins comprising,

means for culturing a glycosylated ~~molecule~~protein-producing cell line;

means for measuring a level of one or more glycan structures, wherein the means generates spectral data, wherein the spectral data comprises Raman spectra;

means for generating one or more regression models based on the spectral data; and

means for measuring a level of one or more glycan structures in a glycosylated ~~molecule~~protein-producing cell line.

11. The system of claim 10, wherein:

(a) the glycosylated ~~molecule~~protein-producing cell line is a mammalian cell line, optionally wherein the mammalian cell line is a non-human cell line;

(b) the culturing comprises batch, fed-batch, perfusion, or~~f~~ a combination thereof;

(c) culturing comprises a volume of:

(i) about 2,000 liters or more;

(ii) about 5,000 liters or more;

(iii) about 10,000 liters or more;

(iv) about 15,000 liters or more;

(v) about 10,000 liters to about 25,000 liters; or

(vi) about 15,000 liters;

(d) measuring is performed in-line, at-line, on-line, off-line, or a combination thereof;

(e) measuring occurs:

(i) more than once daily;

(ii) about every 5 to 60 minutes;

(iii) about every 10 to 30 minutes;

(iv) about every 10 to 20 minutes; or

(v) about every 12.5 minutes; and/or

(f) the one or more glycan structures are selected from the group consisting of: G0F-GlcNac, G0, G0F, G1F, and G2F, or a combination thereof.

12. The system of claim 10 or 11, further comprising a means for selectively modifying one or more operating parameters to enhance production of a desired one or more glycan structures,

wherein the one or more operating parameters comprise a pH level, a nutrient level, a culture media concentration, a frequency interval for culture media addition, or a combination thereof,

wherein the nutrient level is optionally selected from the group consisting of: a concentration of glucose, a concentration of lactate, a concentration of glutamine, and a concentration of ammonium ions, and

wherein the concentration of glucose is optionally automatically modified based on the measured spectral data.

13. The system of any one of claims 10 to 12:

(a) wherein the one or more glycosylated ~~molecules~~proteins comprise a monoclonal antibody (mAb) or a non-mAb;

~~(b) wherein the spectral data comprises Raman spectra;~~

(b~~e~~) wherein the one or more regression models comprise a partial least squares (PLS) model; and/or

(c~~d~~) further comprising a means for isolating the one or more glycosylated ~~molecules~~proteins.

14. A system for producing one or more glycosylated ~~molecules~~proteins, the system comprising:

a bioreactor comprising a glycosylated ~~molecule~~protein-producing cell line;

a process analytical technology (PAT) tool that measures one or more glycan structures and generates spectral data, wherein the PAT tool utilizes or otherwise comprises Raman spectroscopy; and

a processor that correlates levels of one or more glycan structures with the spectral data using one or more regression models.

15. The system of claim 14, wherein:

(a) the bioreactor is:

(i) about 2,000 liters or more;

(ii) about 5,000 liters or more;

(iii) about 10,000 liters or more;

(iv) about 15,000 liters or more;

(v) about 10,000 liters to about 25,000 liters; or

(vi) about 15,000 liters.

(b) the glycan structure is selected from the group consisting of: G0F-GlcNac, G0, G0F, G1F, and G2F, or a combination thereof;

(c) the glycosylated ~~molecule~~protein-producing cell line is a mammalian cell line, optionally wherein the mammalian cell line is a non-human cell line; and/or

~~(d) the PAT tool utilizes or otherwise comprises Raman spectroscopy; and/or~~

(d~~e~~) the one or more regression models comprise a partial least squares (PLS) model.